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*2025 Spring Math*

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HTOP,PDE,COMPLEX,ALGO

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## Chapter 1

# Functions of Complex Variables

### §1.1 Lecture 1 (02-03) Complex numbers

#### §1.1.1 Introduction

You know the sets

$$(R, +, \times) \text{ and } (Q, +, \times)$$

$Q(\sqrt{2}) = \{a + b\sqrt{2} | a, b \in Q\}$   $Q(\sqrt{2})$ , they are infinite sets.

#### Definition 1.1.1: Fields

We say that a set  $F$  together with two binary operations, called addition  $+, F \times F \rightarrow F$ , and multiplication  $\times, F \times F \rightarrow F$  satisfying:

- associativity  $(a + b) + c = a + (b + c)$
- commutativity  $a + b = b + a, ab = ba$
- existence of identities  
there is a  $O \in F$  such that  $a + O = a$  for all  $a \in F$   
there is a  $1 \in F$  such that  $a \times 1 = a$  for all  $a \in F$
- existence of inverses  
for every  $a \in F$  there is a  $b \in F$  such that  $a + b = O$   
for every  $a \in F$  there is a  $b \in F$  such that  $a \times b = 1$
- distributivity  
 $a \times (b + c) = a \times b + a \times c$

#### Example 1: Fields

- For every prime number  $p$ , and  $n \geq 0, n \in N$ , there is a field  $(F_{p^n}, +, \times)$  with  $p^n = q$  elements (cardinality  $q$ ).
- $(C, +, \times)$  is a field

Question:  $(R^d, +), d \geq 2$ , is there a multiplication,  $\times$  such that  $(R^d, +, \times)$  is a field?

- $R^2$ , scalar product,  $x \cdot y = x_1y_1 + x_2y_2 \in R$  is not a field because  $R^2 \times R^2 \rightarrow R$ .
- cross product, not satisfying Definition 1 and 5.
- $d=2$ , there is a good definition of multiplication,  $(R^2, +, \times) \cong (C, +, \times)$ .

**Lemma 1.1.1**

Consider the vector space  $(\mathbb{R}^2, +)$   
 Define the multiplication:  $(a, b) \times (c, d) = (ac - bd, ad + bc)$   
 Then  $(\mathbb{R}^2, +, \times)$  is a field.

Proof:

- You can check that  $\times$  is commutative and satisfies associativity and distributivity.
- We can check  $(1, 0)$  is a multiplication identity.
- also if  $(a, b) \neq (0, 0)$ , then

$$(a, b) \times \left( \frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) = \left( \frac{a^2 + b^2}{a^2 + b^2}, \frac{-ab + ab}{a^2 + b^2} \right) = (1, 0)$$

## §1.2 Lecture 2 (02-05)Complex numbers

### §1.2.1 The algebra of complex numbers

It is important to classify sets:

in Set Theory: bijection

in Topology: homeomorphism

For **Fields**: isomorphism

**Definition 1.2.1: isomorphism**

We say that fields  $F$  and  $F'$  are isomorphic if there exists a function

$$i : F \rightarrow F'$$

which is a bijection and preserve the binary operations

$$i(a + b) = i(a) + i(b)$$

and

$$i(ab) = i(a)i(b)$$

.

$i$  is called an isomorphism and we write  $F \cong F'$

**Comments:**

There is a unique field up to isomorphism of order (its cardinality)

$$p^n : F_{p^n}$$

**Definition 1.2.2: subfield**

Given a field  $(F, +, \times)$ , we say that  $(E, +, \times)$  is a subfield of  $(F, +, \times)$  if  $E \subset F$  and the addition and multiplication of  $E$  is the same addition and multiplication of  $F$ .

**Example 1: subfield**

$$Q \subset R$$

$$Q \subset Q(\sqrt{2}) \subset R \subset R^2$$

**Definition 1.2.3: complex number**

Let  $F$  be a field such that:

- $R$  is a subfield of  $F$
- there is an element  $i \in F$  which is the root of the equation

$$x^2 + 1 = 0 (i^2 = -1)$$

Then we define the field of complex numbers  $C$  as

$$C = \{x \in F : x = \alpha + i\beta, \alpha, \beta \in R\}$$

**Comments:**

- We need to show that there is at least one field  $F$  satisfying these properties.  
**Actually,  $(F, +, \times) = (R, +, \times)$  satisfies these properties.**
- $R' = \{x \in R : x = (\alpha, 0), \alpha \in R\} \cong R \rightarrow R$  is a subfield of  $R^2$ .
- Define  $i := (0, 1) \in R^2$

$$(a, b) \times (c, d) := (ac - bd, ad + bc)$$

Then,

$$i^2 = (0, 1)^2 = (0, 1) \times (0, 1) = (-1, 0)$$

**Example 2: exercise**

Set of  $2 \times 2$  real matrices of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}, a, b \in R$$

This set together with the standard addition and multiplication of matrices is a field and it is isomorphic to  $(R^2, +, \times)$ .

$$F = R^2, C = R^2$$

**Comments:**

It turns out that any definition of  $C$  is unique (up to isomorphisms).

**Lemma 1.2.1**

Let  $F$  and  $F'$  be two fields satisfying the properties of the definition of complex numbers, then call  $\mathbb{C}$  and  $\mathbb{C}'$  the corresponding complex number fields. Then  $\mathbb{C} \cong \mathbb{C}'$ .

**Proof:**

Call  $i \in \mathbb{C}$  and  $i' \in \mathbb{C}'$  the square roots  $-1$ .

Then:

$f : \mathbb{C} \rightarrow \mathbb{C}'$  defined as  $f(\alpha + i\beta) = \alpha + i'\beta$  is an isomorphism

$$z = \alpha + i\beta, z' = \alpha' + i'\beta', w = \gamma + i\delta, w' = \gamma' + i'\delta'$$

**Lemma 1.2.2**

Consider the vector space  $(\mathbb{R}, +)$ . Then any multiplication ' $\times$ ' defined in this vector space which transforms it into a field  $(\mathbb{R}, +, \times)$ , is such that  $(\mathbb{R}, +, \times) \cong \mathbb{C}$ .

**Proof:**

Consider the elements

$$(1, 0) \text{ and } j = (0, 1) \in \mathbb{R}^2$$

Then any element of  $\mathbb{R}^2$  can be written as  $\alpha + j\beta, \alpha, \beta \in \mathbb{R}$

Then

$$j^2 = a + jb, a, b \in \mathbb{R}$$

Then,

$$\begin{aligned} j^2 - jb = a, j^2 - 2\frac{jb}{2} = a &\rightarrow j^2 - 2\frac{jb}{2} + \frac{b^2}{4} = a + \frac{b^2}{4} \\ \left(j - \frac{b}{2}\right)^2 &= a + \frac{b^2}{4} \end{aligned} \tag{1.1}$$

**Define:**  $i = \frac{j - \frac{b}{2}}{\sqrt{a + \frac{b^2}{4}}}$

**Claim:**

$$a + \frac{b^2}{4} < 0$$

If:

$$a + \frac{b^2}{4} \geq 0$$

Then (1.1) can be written as

$$\rightarrow \left(j - \frac{b}{2}\right) + \sqrt{a + \frac{b^2}{4}} \left(j - \frac{b}{2}\right) - \sqrt{a + \frac{b^2}{4}} = 0$$

either

$$j - \frac{b}{2} = \sqrt{a + \frac{b^2}{4}}, j - \frac{b}{2} = -\sqrt{a + \frac{b^2}{4}}$$

## §1.3 Complex numbers

The field of rationals  $\mathcal{Q}$  is the fractional field of the integral domain  $\mathcal{Z}$ .  
With the metric

$$d(x, y) = |x - y|$$

$$|x| = \max\{x, -x\}, \forall x, y \in \mathcal{R}$$

The field  $\mathcal{Q}$  is not a complete metric space.  
Its completion  $\mathbb{R}$ , nevertheless is not algebraically closed.  
For instance,  $f(x) = x^2 + 1$ , doesn't split in  $\mathbb{R}$ .  
So we define the splitting field for  $f(x)$  over  $\mathbb{R}$   
i.e. the smallest field extension of  $\mathbb{R}$  such that  $f(x)$  has roots.

Consider the principal ideal  $\langle f(x) \rangle$  in  $\mathbb{R}[x]$   
since  $f(x)$  is irreducible, then  $\langle f(x) \rangle$  is a maximal ideal.  
Then the quotient ring  $\mathbb{C} = \frac{\mathbb{R}[x]}{\langle f(x) \rangle}$  is a field!  
which is the splitting field of  $f(x)$   
The field  $\mathbb{C}$  has no further splitting extensions! All polynomials in  $\mathbb{C}[x]$  split in  $\mathbb{C}$ .  
So as soon as we forced one polynomial.  $f(x) = x^2 + 1$  to have roots, we in fact imposed that all polynomials have roots.

### Topology:

The Topology of  $\mathbb{C}$  is isomorphic to  $\mathbb{R}^2$ .  
Thus  $\mathbb{C}$  is locally compact, but not compact space.  
The Alexandroff compactification says that we can add a point at infinity to  $\mathbb{C}$  so that  $\mathbb{C} \cup \{\infty\}$  compact.

#### Definition 1.3.1: Alexandroff compactification

- A set  $U \supseteq \mathbb{C} \cup \{\infty\}$  is an open neighborhood of  $\{\infty\}$  if  $U = \mathbb{C} \cup \{\infty\} \setminus K$   
When  $K \supseteq \mathbb{C}$ .

$\mathbb{C} \cup \{\infty\}$  is called the Riemann sphere.

#### Definition 1.3.2: Holomorphic Functions

A function  $f : \Omega \rightarrow \mathbb{C}$  is holomorphic where  $\Omega \subseteq \mathbb{C}$  is holomorphic iff:

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) \text{ exists, } \forall z_0 \in \Omega$$

$$f(z) = f(x, y) = u(x, y) + iv(x, y), \text{ where } z = x + iy \in \Omega$$

$$u(z) = \operatorname{Re}(f(z)), v(z) = \operatorname{Im}(f(z))$$

$f : \Omega \rightarrow \mathbb{C}$  being holomorphic of course implies that  $u, v : \Omega \rightarrow \mathbb{R}$  differentiable.  
If we take  $f(z_0) = z_0 = 0$  and we write

$$\frac{\partial}{\partial z} = \frac{1}{2} \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

Then,

$$\frac{f(x) - f(z_0)}{z - z_0} = \frac{f(z)}{z} = \left(\frac{\partial f}{\partial z}\right)_{(z_0=0)} + \frac{\bar{z}}{z} \left(\frac{\partial f}{\partial \bar{z}}\right)_{(z_0=0)} + o(z)$$

For real  $z \rightarrow 0$ , we have  $\frac{\bar{z}}{z} = 1$ , and for purely imaginary  $z \rightarrow 0$ , we have  $\frac{\bar{z}}{z} = -1$ .  
we have to restrict  $\frac{\partial f}{\partial \bar{z}} = 0$

**Cauchy-Riemann equations:**

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

$$f'(z) = \frac{\partial f}{\partial z} = \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)(x, y), \text{ where } z = x + iy \in \Omega$$

Nevertheless,  $f$  being holomorphic is not equivalent to say  $f$  satisfies the Cauchy-Riemann equations.

Indeed we take  $f(z) = \frac{z^5}{|z|^4}$  on  $\mathbb{C} \setminus \{0\}$  and  $f(0) = 0$ .

We can observe that  $f(z)$  satisfies the Cauchy-Riemann equations on  $\mathbb{C}$  but  $f$  is not even continuous at  $z = 0$ .

**Theorem 1.3.1: Looman Menchoff Theorem**

A continuous function  $f : \Omega \cup \mathbb{C} \rightarrow \mathbb{C}$  is holomorphic iff  $f$  satisfies the Cauchy-Riemann equations.

$$\forall f, g \in H(\Omega)$$

$$(fg) = f'g + fg'$$

$$\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$$

$$(g \circ f)'(z) = g'(f(z)) \cdot f'(z)$$

## Chapter 2

# Honors Theory of Probability

## §2.1 Lecture 1 (02-03) Introduction to Probability

Office Hour: Wed 12:30-1:30pm, Fri 3:30-4:30pm W910

- Homework: weekly
- Grades: 5% participation, 15% homework, 40% midterm, final group project (Max 4 people a group, presentation 20%, Final report 20% 10 pages)

### §2.1.1 Intro

Why do we need modern theory of probability?

#### Example 1

- Coin flip 7 times, what is  $P[\text{first outcome and fifth outcome are Head}]$ ?  
sol:

$$\Omega = (x_1, x_2, x_3, x_4, x_5, x_6, x_7)$$

$$A = (x_1, x_2, x_3, x_4, x_5, x_6, x_7), x_1 = x_5 = H$$

$$P(A) = \frac{|\Omega|}{|A|} = \frac{1}{4}$$

- Stock Price (mathematically, geometric Brownian motion, Stochastic process, such that  $t \rightarrow S_t$  is continuous but nowhere differentiable)  
What is  $P[S_T \geq 100]$ ?

$$\Omega = C[0, T] = \{\text{continuous function}, f[0, T] \rightarrow R\}$$

$$A = \{f \in C[0, T] | f(t) \geq 100\}$$

requires measure theory that defines  $P[S_T \geq 100]$

In modern Prob, Probability Space:  $(\Omega, F, P)$ ,  $\Omega$  is sample space,  $F$  is  $\sigma$ -algebra (meaningful subset of  $\sigma$ ),  $P$  is probability measure ( $P : F \rightarrow [0, 1]$ ).

Topics covered:

- Probability space,  $\sigma$ -algebra, measure, Conditional Probability and Independence
- Random variables (measurable functions), distribution
- Expectation (Lebesgue integral), Conditional distribution and expectation, functions of random variables, Radon-Nikodym Derivative
- Random walks

- generating functions, characteristic functions
- Branching processes
- Convergence of random variables, Law of large numbers, Monte-Carlo Method
- Central Limit Theorem
- Time permitting: Large deviations, Markov Chains

### §2.1.2 Probability Space

$(\Omega, F, P)$

#### Example 2: Coinflip and Stock Price

- Coin flip infinite times,

$$\Omega = \{(x_1, x_2, \dots), x_i = H, T\}$$

Let  $A$  be the event of  $10^6$  consecutive Tails,

$$A = \bigcup_{i=1}^{+\infty} \{x_i = x_{i+1} = \dots = x_{i+10^6-1} = 0, (x_i \in \Omega)\}$$

$$P[A] = 1$$

- Discrete Stock Price model,  $t=0,1,2,\dots,T$   
 $T$  is maturing time, time step  $\Delta t \ll T$

$$N = \frac{T}{\Delta t}$$

$$\text{price go} \begin{cases} \nearrow \text{ by factor : } e^{\sigma\sqrt{\Delta t}} \\ \searrow \text{ by factor : } e^{-\sigma\sqrt{\Delta t}} \end{cases}$$

$$\Omega = \{(x_1, x_2, \dots, x_N), x_i = 0 \text{ or } 1\}$$

Stock price at time  $t$ :  $\forall \omega \in \Omega$

$$S_N(\omega) = S_0 e^{\sum_{i=1}^N x_i \sigma \sqrt{\Delta t} e^{(N - \sum_{i=1}^N x_i)(-\sigma\sqrt{\Delta t})}}$$

$$S : \Omega \rightarrow R(\text{Random Variable})$$

Event: return at  $T$  is positive but not more than 10%:

$$\{\omega \in \Omega : S_N(\omega) \in (S_0, 1.1S_0]\}$$

#### Example 3: Gambling

- Gambling:

- start with  $\{0,1,2,\dots\}$  each time bet an integer amount
  - if amount of money =0, stays at 0
- wealth process:  $\Omega = \{0,1,2,\dots\} \times \{0,1,2,\dots\} \times \dots$   
wealth after time n: (random variable)  $X_n : \Omega \rightarrow N, (x_1, x_2, \dots, x_n) \rightarrow x_n$   
 $(X_n)$  is a Markov Chain (future only depends on present state, but not the past)

$$\text{Event: } \{\text{State } j \text{ is reached from state } i\} \quad (2.1)$$

$$= \{\omega \in \Omega : \exists n \in N, X_0(\omega) = i, X_n(\omega) = j\} \quad (2.2)$$

$$= \bigcup_{m=1}^{+\infty} \{\omega \in \Omega : X_0(\omega) = i, X_m(\omega) = j\} \quad (2.3)$$

## §2.2 Lecture 2 (02-05) – Algebra

### §2.2.1 algebra and $\sigma$ -algebra

in practice, want  $F$  to be closed under  $\bigcap, \bigcup, ^c$

#### Definition 2.2.1

Let  $\mathcal{A}$  be a collection of subsets of  $\Omega$ ,  $\mathcal{A}$  is an algebra iff:

- $\Omega \in \mathcal{A}$
- if  $A, B \in \mathcal{A}$  then  $A \cap B \in \mathcal{A}$
- if  $A \in \mathcal{A}$  then  $A^c \in \mathcal{A}$

#### Remark 1

if  $A, B \in \mathcal{A} \rightarrow A \cap B \in \mathcal{A}$ , because  $A \cap B = (A^c \cup B^c)^c$

**Fact:**

- ①  $P(\Omega)$ (powerset) =  $\{A : A \subset \Omega\}$  is an algebra
- ② smallest algebra/trivial algebra:  $\{\emptyset, \Omega\}$
- ③ Let  $A_1, A_2$  be two algebras of  $\Omega$

$$A_1 \cap A_2 = \{B \in \Omega : B \in A_1 \text{ and } B \in A_2\}$$

if  $(A_j)_{j \in J}$  is a family of algebras, then  $\bigcap_{j \in J} A_j$  is an algebra

- ④ Let  $\mathcal{E}$  be any collection of subsets of  $\Omega$

$$a(\mathcal{E}) = \bigcap_{\substack{A \supseteq \mathcal{E} \\ A \text{ is an algebra}}} A$$

$a(\mathcal{E})$  is an algebra by ③

in fact,  $a(\mathcal{E})$  is the smallest algebra containing  $\mathcal{E}$

It is called the algebra generated by  $\mathcal{E}$

- ⑤ Let  $A \subseteq \Omega, \mathcal{E} = \{A\}$   
Then

$$a(\mathcal{E}) = \underbrace{\{A, A^c, \Omega, \emptyset\}}_f$$

**Proof:**

- $a(\mathcal{E}) \subseteq f$   
notice that  $f$  is an algebra since  $f \supseteq \mathcal{E}$ , therefore  $f \supseteq a(\mathcal{E})$  because  $a(\mathcal{E})$  is the smallest algebra containing  $\mathcal{E}$ .
- $f \subseteq a(\mathcal{E})$ :

$$A \in a(\mathcal{E}), A^c \in a(\mathcal{E})$$

because  $a(\mathcal{E})$  is an algebra,  $\emptyset, \Omega \in a(\mathcal{E})$

- ⑥  $\pi = \{A_1, A_2, \dots, A_n\}, \Omega = \bigcup_{i=1}^n A_i, A_i \cap A_j = \emptyset$  Then:

$$a(\pi) = \left\{ \bigcup_{i \in I} A_i, \text{ for } I \subset 1, 2, \dots, n \right\} = \text{finite disjoint union of } (A_i)_{i=1}^n$$

- ⑦ Let  $\mathcal{A}$  be an algebra of  $\mathbb{R}$   
Let  $X : \Omega \rightarrow \mathbb{R}$  be a function  
Then

$$\underbrace{\{X^{-1}(A), A \in \mathcal{A}\}}_{\{\omega \in \Omega : X(\omega) \in A, \text{ for some } A \in \mathcal{A}\}} \text{ is an algebra of } \Omega$$

**Hint:**

$$X^{-1}(A \cup B) = (X^{-1}(A)) \cup (X^{-1}(B))$$

- ⑧  $\Omega = \mathbb{R}, \mathcal{E} = \{\text{left open right closed intervals}\} = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty), & -\infty \leq a \end{cases}$

Then:

$$\begin{aligned} a(\mathcal{E}) &= \text{"finite disjoint union of elements in } \mathcal{E}\text{"} \\ &= \underbrace{\{I_1 \cup \dots \cup I_k, I_j \in \mathcal{E}, I_j \cap I_j = \emptyset(f)\}}_f \end{aligned}$$

**Hint:**

- $f \subseteq a(\mathcal{E})$  straightforward
- $a(\mathbb{R}) \subseteq f$ 
  - check  $f$  is an algebra
  - since  $\mathcal{E} \subset f$ ,  $a(\mathcal{E})$  is the smallest algebra containing  $\mathcal{E}$ ,  $a(\mathcal{E}) \subseteq f$
  - $(a, b]^c = \underbrace{(-\infty, a] \cup (b, +\infty)}_{\in f}$

### Lemma 2.2.1

In Probability, if  $(A_k)_{k=1}^n$  are disjoint events, we have  $P(\bigcup_{k=1}^{+\infty} A_k) = \sum_{k=1}^{+\infty} P(A_k)$

**Definition 2.2.2:**  $\sigma$  – algebra

A  $\sigma$  – algebra ( $\sigma$  – field)  $\mathcal{A}$  is a collection of subsets of  $\Omega$  such that

- $\Omega \in \mathcal{A}$
- if  $A_1, A_2, \dots \in \mathcal{A}$  then  $\bigcup_{i=1}^{+\infty} A_i \in \mathcal{A}$
- if  $A \in \mathcal{A}$ , then  $A^c \in \mathcal{A}$

**Remark 2**

if  $A_1, A_2, \dots \in \mathcal{A}$  then  $\bigcap_{i=1}^{+\infty} A_i \in \mathcal{A}$

$\sigma$  – algebra represents the collection of information determined by partial derivatives

**Example 1**

- coinflips infinity many times

$$\Omega = \{(x_1, x_2, \dots) | x_i = 0, 1\} = (0, 1)^\infty$$

After observing first outcome,  $\mathcal{F}_1 = \{\emptyset, \Omega, A_0, A_1\}$

Where  $A_0 = \{(0, x_2, x_3, \dots) | x_i = 0, 1\}$ ,  $A_1 = \{(1, x_2, x_3, \dots) | x_i = 0, 1\}$

After observing second outcome,  $\mathcal{F}_2 = \{\emptyset, \Omega, A_0, A_1, A_00, A_01, A_10, A_11\}$

Where  $A_{00} = \{(0, 0, x_3, x_4, \dots) | x_i = 0, 1\}$ ,  $A_{01} = \{(0, 1, x_3, x_4, \dots) | x_i = 0, 1\} \dots$

After observing first  $n$  outcomes,

$$\mathcal{F}_n = \{\emptyset, \Omega, (A_j)_{j \in (\sigma 1)^n} \text{ and finite disjoint unions}\}$$

and

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_n \subseteq \dots$$

**Proposition 2.2.0**

- ① intersection of  $\sigma$  – algebras is a  $\sigma$  – algebra
- ② Let  $\mathcal{E}$  be a collection of subsets of  $\Omega$

$$\sigma(\mathcal{E}) = \bigcap_{\substack{f \supseteq \mathcal{E} \\ f \text{ is an algebra}}} f \text{ is a smallest } \sigma \text{ – algebra that contains } \mathcal{E}$$

- ③ For any collection  $\mathcal{E}$ , we have  $\sigma(\mathcal{E}) \subseteq \sigma(\mathcal{E})$

- ④ For any collection  $\mathcal{E}$ , we have  $\sigma(a(\mathcal{E})) = \sigma(\mathcal{E})$

**Hint:**

$$a(\mathcal{E}) \subseteq \sigma(\mathcal{E}) \rightarrow \sigma(a(\mathcal{E})) \subseteq \sigma(\mathcal{E})$$

$$\mathcal{E} \subseteq \sigma(a(\mathcal{E})) \rightarrow \sigma(\mathcal{E}) \subseteq \sigma(a(\mathcal{E}))$$

## §2.3 Recitation 1 (02-07) – Problem Solving

### §2.3.1 Basic Set Theory

**Definition 2.3.1:**  $\cup, \cap, ^c$

**De Morgan's Law:**

$$\left(\bigcup_{j \in J} A_j\right)^c = \bigcap_{j \in J} A_j^c; \left(\bigcap_{j \in J} A_j\right)^c = \bigcup_{j \in J} A_j^c$$

**Definition 2.3.2:**  $\setminus$

$$A \setminus B = A \cap B^c$$

$$\text{Then } A \setminus B = A \cap B^c = A \setminus (A \cap B) = B^c \setminus A^c$$

**Remark 1**

$$\bigcap_{n \geq 1} A_n = A_1 \setminus \left(\bigcup_{j \geq 2} (A_1 - A_j)\right) \text{ (ex)}$$

### §2.3.2 Limits of Sets

**Definition 2.3.3: Limit Sets**

Let  $(A_n)_{n \geq 1}$  be a sequence of sets, then

$$B_k = \bigcup_{n \geq k} A_n, C_k = \bigcap_{n \geq k} A_n$$

Then  $B_k$  is increasing,  $C_k$  is decreasing

Define:

$$\limsup_{n \rightarrow +\infty} A_n = \lim_{k \rightarrow +\infty} B_k = \bigcap_{k \geq 1} \bigcup_{n \geq k} A_n$$

$$\liminf_{n \rightarrow +\infty} A_n = \lim_{k \rightarrow +\infty} C_k = \bigcup_{k \geq 1} \bigcap_{n \geq k} A_n$$

**Definition 2.3.4: liminf, limsup of sequence**

$$\limsup a_n = \inf_{k \geq 1} \sup_{n \geq k} a_n, \liminf a_n = \sup_{k \geq 1} \inf_{n \geq k} a_n$$

When  $\limsup A_n = \liminf A_n$ , Then we say  $\lim_{n \rightarrow +\infty} A_n$  exist and  $\lim_{n \rightarrow +\infty} A_n = \limsup A_n = \liminf A_n$

In probability,

$$\begin{aligned} \limsup A_n &= \{A_n \text{ occurs infinitely often}\} \\ &= \{A_n.i.o\} \\ x \in \limsup A_n &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k \text{ such that } x \in A_n \\ &\Leftrightarrow A_n.i.o \\ \liminf A_n &= \{A_n \text{ occurs eventually}\} \\ &\Leftrightarrow \exists k \in \mathbb{N}, \forall n \geq k, x \in A_n \end{aligned}$$

**Remark 2**

1. if  $A_n$  increases, then  $\limsup A_n = \lim_{n \rightarrow \infty} A_n = \bigcup_{n \geq 1} A_n$   
if  $A_n$  decreases, then  $\liminf A_n = \lim_{n \rightarrow \infty} A_n = \bigcap_{n \geq 1} A_n$

2.

$$(\limsup A_n)^c = \liminf A_n, (\liminf A_n)^c = \limsup A_n$$

**§2.3.3 Exercise**

1.  $A_k = \begin{cases} E, & \text{if } k \text{ is odd} \\ F, & \text{if } k \text{ is even} \end{cases}$

Then  $\limsup A_n = E \cup F, \liminf A_n = E \cap F$

2. Let  $f_n : \mathbb{R} \rightarrow \mathbb{R}$ , Let  $A = \{x \in \mathbb{R} : \lim_{n \rightarrow \infty} f_n(x) = f(x)\}$  Then,

$$\begin{aligned} A^c &= \bigcup_{k=1}^{+\infty} \bigcap_{m=1}^{+\infty} \bigcup_{n=m}^{+\infty} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\} \\ &= \bigcup_{k=1}^{+\infty} [\limsup_n \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}] \end{aligned}$$

3. Suppose that  $\lim_{n \rightarrow \infty} f_n(x) = f(x), \forall x \in \mathbb{R}$   
Then

$$\begin{aligned} \{x : f(x) \leq t\} &= \bigcap_{k=1}^{+\infty} \bigcup_{m=1}^{+\infty} \bigcup_{n=m}^{+\infty} \{x \in \mathbb{R} : f_n(x) < t + \frac{1}{k}\} \\ &= \bigcap_{k=1}^{+\infty} [\liminf_n \{x : f_n(x) < t + \frac{1}{k}\}] \end{aligned}$$

**Proof:**

1. ex2: Want

$$A = \bigcup_{k \geq 1} \bigcap_{n \geq 1} \bigcup_{m \geq n} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}$$

$$f_n(x) \rightarrow f(x) \text{ iff } \forall \epsilon > 0, \exists N \geq N, |f_n(x) - f(x)| < \epsilon$$

$$\{x : f_n(x) \rightarrow f(x)\} = \bigcap_{\epsilon > 0} \bigcup_N \bigcap_{n \geq N} \{x : |f_n(x) - f(x)| < \epsilon\},$$

USE that  $\{|f_n - f| < \epsilon\}$  is monotone increasing in  $n$

**Review on mapping:**

$$f : X \rightarrow Y, f^{-1} : Y \rightarrow X$$

**Basic properties:**

- $f(\bigcup_{i \in I} A_i) = \bigcup_{i \in I} f(A_i)$
- $f(\bigcap_{i \in I} A_i) = \bigcap_{i \in I} f(A_i)$

For  $(B_i)$  subset of  $Y$ :

- if  $B_1 \subset B_2, f^{-1}(B_1) \subset f^{-1}(B_2)$
- $f^{-1}(\bigcup_{i \in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i)$
- $f^{-1}(B^c) = (f^{-1}(B))^c$

### Definition 2.3.5: Indicator Mapping

$$1_A : X \rightarrow \{0, 1\}$$

$$1_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

**Exercise:**

1.  $1_{\limsup A_n} = \limsup 1_{A_n}$
2.  $1_{\liminf A_n} = \liminf 1_{A_n}$

**Proof:** if

$$\begin{aligned} 1_{\limsup A_n}(x) = 1 &\Leftrightarrow x \in \limsup A_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, x \in A_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, 1_{A_n}(x) = 1 \\ &\Leftrightarrow \limsup 1_{A_n}(x) \geq 1 \text{ (Definition of limsup)} \\ &\Leftrightarrow 1_{\limsup A_n}(x) = 1 \end{aligned}$$

**Exercise:** Let  $\mathcal{A}_1, \mathcal{A}_2$  be algebras of  $\Omega$

1. show that  $\underbrace{\mathcal{A}_1 \cap \mathcal{A}_2}_{B \subset \Omega: B \in \mathcal{A}_1, B \in \mathcal{A}_2}$  is an algebra

2. show that  $\underbrace{\mathcal{A}_1 \cup \mathcal{A}_2}_{B \subset \Omega: B \in \mathcal{A}_1 \text{ or } B \in \mathcal{A}_2}$  is an algebra iff  $\mathcal{A}_1 \subseteq \mathcal{A}_2$  or  $\mathcal{A}_2 \subseteq \mathcal{A}_1$

**Proof:** Suppose by contradiction that

$$\exists A_1 \in \mathcal{A}_1 \text{ but } A_1 \notin \mathcal{A}_2 \text{ and } A_2 \in \mathcal{A}_2 \text{ but } A_2 \notin \mathcal{A}_1 \text{ and } \mathcal{A}_1 \cup \mathcal{A}_2 \text{ is an algebra}$$

Therefore:

$$A_1 \cup A_2 \in \mathcal{A}_1 \cup \mathcal{A}_2, A_1 \setminus A_2 = A_1 \cap \underbrace{A_2^c}_{\mathcal{A}_1 \cup \mathcal{A}_2} \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$$A_2 \setminus A_1 = A_2 \cap A_1^c \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$\Rightarrow$  at least two of

$$(A_1 \setminus A_2) \cup (A_2 \setminus A_1), A_1 \setminus A_2, A_2 \setminus A_1 \text{ are in } \mathcal{A}_1 \text{ or } \mathcal{A}_2$$

**Assume in  $\mathcal{A}_1$**

$\Rightarrow$  (by  $\mathcal{A}_1$  is an algebra, all three sets are in  $\mathcal{A}_1$ )

$$\Rightarrow A_2 = \underbrace{(A_1 \cup A_2)}_{\in \mathcal{A}_1} \setminus \underbrace{(A_1 \setminus A_2)}_{\mathcal{A}_1} \in \mathcal{A}_1$$

**Contradiction!**

## Chapter 3

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# Partial Differential Equations

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### §3.1 Lecture 1 (02-03) Initial and Boundary Conditions

#### §3.1.1 What are the PDEs?

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - u \quad (3.1)$$

$$u = u(x, y, t) \quad (3.2)$$

(3.1) is a second order PDE

#### §3.1.2 Classical solutions

To simplify, we consider here equations of one unknown function  $u$  of one or more variable

By a solution we mean a sufficiently smooth function  $u$  that satisfies the PDE at every point of the domain of its definition( $D$ ).

Sometimes, we will have to impose some boundary conditions on the domain  $D$ .

$C^n = \{u : D \rightarrow R : n\text{-times continuous differentiable w.r.t all variables} \}$   $n$  is the order of the equation

#### Example 1

- Heat equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$$u = u(x, t)$$

$$(t, x) \in D \subset R^2$$

$$u(t, x), \frac{\partial u}{\partial t}, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial t^2}$$

$$n = 2$$

$$\frac{\partial^2 u}{\partial x \partial t} = \frac{\partial^2 u}{\partial t \partial x} \in C^2$$

$$u \in C^2(D)$$

- Some Solution:

$$u(t, x) = t + \frac{x^2}{2}, D = R^2$$

$$u(t, x) = \frac{e^{-\frac{x^2}{4t}}}{2\sqrt{\pi t}}, D = \{(t, x) : t > 0\}$$

$$u(t, x) = e^{-t+ix} = e^{-t}\cos(x) + ie^{-t}\sin(x), D = \mathbb{R}^2$$

### §3.1.3 Initial and Boundary Conditions

What is Initial data?

$$u(0, x) = u_0(x)$$

What are Boundary Conditions? rigorous definition of Boundary

- Dirichlet Boundary Condition:  $u(t, x)|_{\partial D} = g(t, x)$
- Neumann Boundary Condition: When the normal derivative of  $u$  is specified  $\frac{\partial u}{\partial n}$  where  $n$  is the normal to the boundary  $\partial(D)$

$$\frac{\partial u}{\partial n}(t, x) = g(t, x)$$

- Mixed Boundary Condition: Combination of Dirichlet and Neumann

### §3.1.4 Linear and Nonlinear waves

**Stationary waves**

$$\frac{\partial u}{\partial t} = 0 \tag{3.3}$$

$$u = u(t, x) \tag{3.4}$$

This is a first order linear homogeneous PDE

Solution of the form  $u = u(t) = c, \forall c \in \mathbb{R}$

Let's integral (3.3) :

$$0 = \int_0^t \frac{\partial u}{\partial t}(s, x) ds = u(t, x) - u(0, x)$$

Thus if an initial conditional

$$u(0, x) = f(x)$$

$$u(t, x) = f(x), \forall t, x$$

Thus for any given  $f \in C^1(\mathbb{R})$ , we have a solution of (3.3) and this is a stationary wave

The domain of this solution is  $\mathbb{R}$  if  $f$  is defined everywhere on  $\mathbb{R}$

Power of domain to change the solution:

Consider the ODE:

$$\frac{du}{dt} = 0, D = (-\infty, 0) \cup (0, \infty)$$

The solution is:

$$u(t) = \begin{cases} c_1, t < 0 \\ c_2, t > 0 \end{cases} \quad \forall c_1, c_2 \in \mathbb{R}$$

similarly, for the PDE in example (3.3):

$$u(t, x) = \begin{cases} 0, x > 0 \\ x^2, x \leq 0, t > 0 \\ -x^2, x \leq 0, t < 0 \end{cases}$$

it is a  $\mathbb{C}^1$  solution of (3.3) on  $\mathbb{R}^2$  and it is a classical solution.

## §3.2 Lecture 1 (02-05) Transport and Traveling waves

### §3.2.1 uniform transport

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \quad (3.5)$$

$$u = u(0, x) = f(x) \quad (3.6)$$

$$x \in \mathbb{R}, t_0 = 0, t - t_0 = t, c \in \mathbb{R}$$

use change of variable:

$$\xi = x - ct$$

$\xi$  is called characteristic variable

let's look for a solution of (3.5) under the form

$$u(t, x) = v(t, \xi) = v(t, x - ct)$$

use chain rule:

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} - c \frac{\partial v}{\partial \xi}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial \xi}$$

Thus:

$$0 = \frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = \frac{\partial v}{\partial t} - c \frac{\partial v}{\partial \xi} + c \frac{\partial v}{\partial \xi} = \frac{\partial v}{\partial t}$$

Conclusion:

$$\frac{\partial v}{\partial t} = 0$$

Solution is constant  $v(\xi)$

$$u(t, x) = v(\xi) = v(x - ct)$$

#### Proposition 3.2.0

if  $u$  solves (3.5), then  $u$  has the form  $u(t, x) = v(x - ct)$  where  $v$  is a function of one variable that is  $\mathbb{C}^1$ -smooth.

Actually,  $v$  is satisfying

$$u(0, x) = v(x) = f(x)$$

Conclusion:

$$\begin{cases} \frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \\ u = u(0, x) = f(x) \end{cases}$$

The solution is

$$u(t, x) = f(x - ct)$$

$f(x - ct)$  is a traveling wave solution moving at velocity  $c$ :

if  $c > 0$ , the wave moves to the right

if  $c < 0$ , the wave moves to the left

if  $c = 0$ , the wave is stationary

**Example 1**

$$\begin{cases} \frac{\partial u}{\partial t} + 2\frac{\partial u}{\partial x} = 0 \\ u = u(0, x) = \frac{1}{1+x^2} \end{cases}$$

The solution is

$$u(t, x) = \frac{1}{1 + (x - 2t)^2}$$

The lines  $x = ct + k$  are called the characteristic lines for the equation (3.5) on the characteristic lines, the solution is always constant on geometric interpretation,  $u(t, x) = f(x - ct) = f(k)$

Consider the equation:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} + au = 0 \quad (3.7)$$

where  $a$  and  $c$  are given numbers.

**Use change of variable:**

$$u(t, x) = v(t, \xi) = v(t, x - ct)$$

we will get by the same computation

$$\frac{\partial v}{\partial t} + av = 0$$

The solution will be

$$\begin{aligned} v(\xi) &= e^{-at} f(\xi) \\ u(t, x) &= v(t, \xi) = v(x - ct) = e^{-at} f(x - ct) \end{aligned}$$

So,

$$\begin{cases} u(t, x) = e^{-at} f(x - ct) \\ u(0, x) = f(x) \end{cases}$$

### §3.2.2 Nonuniform transport

$$\frac{\partial u}{\partial t} + c(x) \frac{\partial u}{\partial x} = 0 \quad (3.8)$$

In the uniform case ( $c$  constant), characteristic line is  $x = ct + k$  (the solution of ODE below)

$$\frac{dx}{dt} = c$$

In the general non-uniform case, we can still define characteristic curves

$$\frac{d}{dt} x(t) = c(x(t))$$

Let's show that the solutions of (3.8) are constants along the characteristic curves  $(t, x(t))$

We need to check that  $u(t, x(t)) = h(t)$  is constant of  $t$   
compute the derivative of  $h(t)$

$$\begin{aligned}\frac{d}{dt}h(t) &= \frac{d}{dt}u(t, x(t)) = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} \\ &= \frac{\partial u}{\partial t} + c(x(t)) \cdot \frac{\partial u}{\partial x} = 0\end{aligned}$$

$h(t)$  and thus  $u(t, x(t))$  is constant along the characteristic curves

**Definition 3.2.1**

The graph of a solution of the ODE

$$\frac{d}{dt}x(t) = c(x(t))$$

is called a characteristic curve for the transport equation (3.8) with speed  $c(x)$

Thus at each point  $(t, x)$  the slope of the characteristic curve is  $c(x)$  (the wave speed)

In the particular case  $c(x)=c$ , the characteristic curves are straight lines of slope  $c$

Moreover,

(3.8) is an autonomous first order ODE which can be "solved" using the method of separation of variables

Assume that  $c(x) \neq 0$

$$\begin{aligned}\frac{dx(t)}{dt} &= c(x(t)) \\ \frac{dx}{c(X)} &= dt \rightarrow \\ \beta(x) &= \int \frac{dx}{c(x)} = t + k\end{aligned}$$

Thus characteristic curves satisfies the implicit equation

$$\beta(x) = t + k$$

We can find  $x$  from the equality using  $\beta^{-1}$  (the inverse of  $\beta$ )

$$\begin{aligned}\beta(x) &= t + k \\ \beta^{-1}\beta(y) &= y \\ x(t) &= \beta^{-1}(t + k)\end{aligned}$$

**Remark 1**

How to determine where  $c(x_0) = 0$ ?

Solve this ODE

$$\begin{cases} \frac{dx}{dt} = c(x) \\ x(t_0) = x_0 \end{cases}$$

and  $x(t) = x_0$  will be the characteristic curve going through  $(t_0, x_0)$

From the proposition, we know that  $u(t, x(t)) = \text{constant}$

$$\begin{aligned} u(t, x(t)) &= v(k) \\ &= v(\beta(x(t)) - t) \end{aligned}$$

where  $k$  is the constant determining the characteristic curve  $(t, x(t))$

$$\begin{aligned} u(t, x) &= v(\xi) \\ \text{where } \xi &= \beta(x) - t \end{aligned}$$

which we will call characteristic variable

$$u(t, x) = v(\beta(x) - t) \tag{3.9}$$

This formula may fail at points where  $c$  vanishes

$$\begin{aligned} t &= 0 \\ u(0, x) &= f(x) = v(\beta(x) - 0) \\ f(x) &= v(\beta(x)) \\ v(\xi) &= f(\beta^{-1}(\xi)) \\ &= f \cdot \beta^{-1}(\xi) \\ u(t, x) &= f \cdot \beta^{-1}(\beta(x) - t) \end{aligned}$$

This is the formula for the solution of (3.8) in the general case which is:

$$\begin{cases} \frac{\partial u}{\partial t} + c(x) \frac{\partial u}{\partial x} = 0 \\ u(0, x) = f(x) \end{cases}$$

## Chapter 4

# Algorithm

### §4.1 Lecture 1 (02-03) Asymptotic bounds and big O

#### Definition 4.1.1: big-Theta notation(=)

Let  $f, g : N \rightarrow R^+$ , we say that  $f(n) = \Theta(g(n))$  if there  $\exists a, b > 0$  and  $n_0 \in \mathbb{N}$  such that  $a \cdot g(n) \leq f(n) \leq b \cdot g(n)$  for all  $n > n_0$ .

Intuitively,  $f(n) = \Theta(g(n))$  means that  $f(n)$  grows as fast as  $g(n)$  up to a constant when  $n$  tends to infinity.

Examples:  $3n = \Theta(n)$ ,  $2n^2 + 7n = \Theta(n^2)$ ,  $\ln n^2 = \Theta(\log n)$

#### Definition 4.1.2: big-O notation( $\leq$ )

Let  $f, g : N \rightarrow R^+$ , we say that  $f(n) = O(g(n))$  if  $\exists n_0 \in \mathbb{N}$  and  $b > 0$  such that  $f(n) \leq b \cdot g(n)$  for all  $n > n_0$ .  
 $\lim_{n \rightarrow \infty} \sup \frac{f(n)}{g(n)} < +\infty$

#### Definition 4.1.3: big-Omega notation( $\geq$ )

Let  $f, g : N \rightarrow R^+$ , we say that  $f(n) = \Omega(g(n))$  if  $\exists n_0 \in \mathbb{N}$  and  $a > 0$  such that  $a \cdot g(n) \leq f(n)$  for all  $n > n_0$ .  
 $\lim_{n \rightarrow \infty} \inf \frac{f(n)}{g(n)} > 0$

#### Remark 1

$$f(n) = \Theta(g(n)) \Leftrightarrow f(n) = O(g(n)) \text{ and } f(n) = \Omega(g(n))$$
$$f(n) = O(g(n)) \Leftrightarrow g(n) = \Omega(f(n))$$

#### Definition 4.1.4: little-o notation( $<$ )

We write  $f(n) = o(g(n))$  if for any  $b > 0$  there  $\exists n_0 \in \mathbb{N}$  such that  $f(n) < b \cdot g(n)$  for all  $n > n_0$ .

**Definition 4.1.5: little-omega notation( $>$ )**

We write  $f(n) = \omega(g(n))$  if for any  $a > 0$  there  $\exists n_0 \in \mathbb{N}$  such that  $a \cdot g(n) < f(n)$  for all  $n > n_0$ .

**Remark 2**

$f(n) = o(g(n)) \Rightarrow f(n) = O(g(n))$  and  $f(n) \neq \Theta(g(n))$   
 $f(n) = \omega(g(n)) \Rightarrow f(n) = \Omega(g(n))$  and  $f(n) \neq \Theta(g(n))$   
 $f(n) = \omega(g(n)) \Leftrightarrow g(n) = \omega(f(n))$

$$n^{1+e}, n \log(n), e \rightarrow 0$$

**Theorem 4.1.1**

Any Algorithm has to check at least  $\log(n+1)$  entries of the sorted array for searching an element in the worst case

**§4.2 Recitation 1 (02-07)–Problem Solving**

| Notation              | Bounding Conditions                                                                       | Relation | Type     |
|-----------------------|-------------------------------------------------------------------------------------------|----------|----------|
| $T(n) = O(f(n))$      | $\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \leq cf(n)$                         | $\leq$   | $O$      |
| $T(n) = \Omega(f(n))$ | $\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \geq cf(n)$                         | $\geq$   | $\Omega$ |
| $T(n) = \Theta(f(n))$ | $\exists c_1, c_2 > 0, \exists n_0, \forall n \geq n_0, c_1 f(n) \leq T(n) \leq c_2 f(n)$ | $=$      | $\Theta$ |
| $T(n) = o(f(n))$      | $\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) < cf(n)$                            | $<$      | $o$      |
| $T(n) = \omega(f(n))$ | $\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) > cf(n)$                            | $>$      | $\omega$ |

**Possible useful formula:**

$$\int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$

1. Multiplication Rule. Prove that for  $h(n) = \Omega(1)$ , if  $f(n) = \Theta(g(n))$  then  $f(n) \cdot h(n) = \Theta(g(n) \cdot h(n))$  (The same also holds for  $O, \Omega, \omega, \omega$ )

**Solution:**

Since  $f(n) = \Theta(g(n))$ ,  $\exists c_1, c_2 > 0, \exists n_0$  such that

$$c_1 g(n) \leq f(n) \leq c_2 g(n), \forall n \geq n_0$$

Since  $h(n) = \Omega(1)$ , there exist constants  $c_3 > 0$  and  $n_1$  such that

$$h(n) \geq c_3, \forall n \geq n_1$$

Combining the two, we have:

$$c_1 g(n) h(n) \leq f(n) h(n) \leq c_2 g(n) h(n)$$

$$\text{Therefore } f(n) h(n) = \Theta(g(n) h(n))$$

2. Addition Rule. Let  $f_1, \dots, f_k : N \rightarrow R_{\geq 0}$  for a fixed number  $k$  and  $f(n) = \sum_{i=1}^k f_i(n)$ . Prove that if  $f_i(n) = O(f_1(n))$  for all  $i \in \{1, \dots, k\}$ , then  $f(n) = \Theta(f_1(n))$ . Here

$f_1$  can be replaced with any one of  $f_1, \dots, f_k$

**Solution:**

Since  $f_i(n) = O(f_1(n))$ , there exist constants  $c_i$  and  $n_0$  such that

$$f_i(n) \leq c_i f_1(n), \forall n \geq n_0$$

Therefore,

$$\sum_{i=1}^k f_i(n) \leq \sum_{i=1}^k c_i f_1(n) = C f_1(n) \text{ where } C = \sum_{i=1}^k c_i$$

Additionally, there exists at least one  $f_j(n) = \Omega(f_1(n))$ , so  $\sum_{i=1}^k f_i(n) = \Theta(f_1(n))$

3. Let  $f$  and  $g$  be non-negative functions. Define  $T(n) = \max(f(n), g(n))$ . Prove that  $T(n) = \Theta(f(n) + g(n))$

**Solution:**

Since  $\max(f(n), g(n)) \leq f(n) + g(n)$ , we have  $T(n) = O(f(n) + g(n))$

Also,  $f(n) + g(n) \leq 2\max(f(n), g(n))$ , so  $T(n) = \Omega(f(n) + g(n))$

Therefore,  $T(n) = \Theta(f(n) + g(n))$

4. Let  $f$  and  $g$  be non-decreasing real-valued functions defined on positive integers and  $f(1) \geq 1$  and  $g(1) \geq 1$ . Prove or disprove that  $f(n) = O(g(n)) \Rightarrow 2^{f(n)} = O(2^{g(n)})$

**Solution:**

Since  $f(n) = O(g(n))$ , there exists a constant  $c > 0$  and  $n_0$  such that

$$f(n) \leq c \cdot g(n), \forall n \geq n_0$$

Therefore,

$$2^{f(n)} \leq 2^{c \cdot g(n)}$$

Since  $2^c$  is constant, we conclude that  $2^{f(n)} = O(2^{g(n)})$

5. Prove that  $\log(n!) = \Theta(n \log n)$ . Hint: To show an upper bound, compare  $n!$  with  $n^n$ . To show a lower bound, compare it with  $\left(\frac{n}{2}\right)^{\frac{n}{2}}$

**Solution:**

We know that:

$$\log(n!) = \log(1) + \log(2) + \dots + \log(n)$$

Since  $\log(n)$  is increasing, the sum can be bounded by:

$$\log(n!) \leq \sum_{k=1}^n \log(n) = n \log(n)$$

From the bound, we can use the fact that  $\log(k) \geq \log\left(\frac{n}{2}\right)$  for all  $k \geq \frac{n}{2}$

Hence, we can sum the lower half of the terms to get:

$$\log(n!) \geq \sum_{k=\frac{n}{2}}^n \log(k) \geq \sum_{k=\frac{n}{2}}^n \log\left(\frac{n}{2}\right) = \frac{n}{2} \log\left(\frac{n}{2}\right)$$

6. Show that  $c^n = o(n!)$  for any constant  $C$

**Solution:**

We need to prove that

$$\lim_{n \rightarrow \infty} \frac{c^n}{n!} = 0$$

Using the recursive relation:

$$\frac{c^n}{n!} = \frac{c^{n-1}}{(n-1)!} \times \frac{c}{n}$$

As  $n \rightarrow \infty$ ,  $\frac{c}{n} \rightarrow 0$ , Thus,

$$\lim_{n \rightarrow \infty} \frac{c^n}{n!} = 0$$

7. Show that  $c^{n^{1.001}} = \omega(n!)$  for any constant  $C$ . Hint: For any fixed numbers  $p, q > 0$ , we have  $n^p = \omega(\log^q n)$

**Solution:**

We need to prove that

$$\lim_{n \rightarrow \infty} \frac{c^{n^{1.001}}}{n!} = \infty$$

Taking the algorithm, we have:

$$\log\left(\frac{c^{n^{1.001}}}{n!}\right) = n^{1.001} \log(c) - \log(n!)$$

Since  $\log(n!) \approx n \log n$ , and  $n^{1.001}$  grows faster than  $n \log n$ , we conclude that

$$\lim_{n \rightarrow \infty} \log\left(\frac{c^{n^{1.001}}}{n!}\right) = \infty$$

$$\text{So, } \lim_{n \rightarrow \infty} \frac{c^{n^{1.001}}}{n!} = \infty$$

Therefore,  $c^{n^{1.001}} = \omega(n!)$

8. Show  $\sum_{i=1}^n i^k = \Theta(n^{k+1})$ . Hint: Show that  $n^{k+1} - (n-1)^{k+1} = \Theta(n^k)$

**Solution:**

Upperbound: Since  $i^k \leq n^k$ , we have

$$\sum_{i=1}^n i^k \leq \sum_{i=1}^n n^k = n^{k+1}$$

Lowerbound: Using the integral approximation

$$\sum_{i=1}^n i^k \geq \int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$

Therefore,

$$\sum_{i=1}^n i^k = \Theta(n^{k+1})$$

9. Search for the peak. Suppose we have an array  $A[1..n]$  and we know that there exists some  $k$  (value unknown) such that  $A[1] < A[2] < \dots < A[k]$  and  $A[k] > A[k+1] > \dots > A[n]$ . We call  $A[k]$  the peak of  $A$ . Design an algorithm to find the peak and analyze it in  $O(\log n)$  time.

**Solution:**

Use Binary Search Algorithm:

1. Set  $l = 1, r = n$

2. While  $l < r$  do

3. Set  $m = \left\lfloor \frac{l+r}{2} \right\rfloor$

4. If  $A[m] < A[m+1]$  then  $l = m + 1$

5. Else  $r = m$

6. Return  $l$

Analysis:

Let  $T(n)$  be the time complexity of the algorithm

Let  $n_0 = 1$

Let  $c = 1$

Let  $T(n) = T\left(\frac{n}{2}\right) + c$

Let  $T(n) = O(\log n)$

10. 3SUM in a sorted array. Let  $A[1 \dots n]$  be a sorted array. We want to see  $A[i] + A[j] + A[k] = x$  for some different  $i, j, k$ , where  $x$  is a given number. Design an algorithm  $3SUM(n, a, x)$  which can return  $(i, j, k)$  or 0 if no such three elements exist. How fast can this problem be solved?

**Solution:**

fix one element  $A[i]$ , then use two pointers on the remaining array to find  $A[j] + A[k] = x - A[i]$

The time complexity is  $O(n^2)$

**Algorithm 3SUM(n,a,x):**

Input: Array A of n integers, integer x

Output: indices i,j,k such that  $A[i] + A[j] + A[k] = x$  or return "No solution" if no such triplet exists  
for i=0 to n-1 do

//Avoid duplicates by skipping repeated elements

if i>0 and  $A[i] == A[i-1]$

continue

left=i+1,right=n-1

while left<right do

sum= $A[i] + A[left] + A[right]$

if sum==x

return i,left,right //Found the triplet

if sum<x

left=left+1 //increase the sum by moving left pointer to the right

else

right=right-1 //decrease the sum by moving right pointer to the left

return "No solution" //no such triplet exists